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BIO 203 — Past Questions & Materials

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Course: BIO 203

Institution: Osun State University, Osogbo

Papers in this pack: 1

Levels: 200

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Paper 1 of 1 — BIO 203&207 PQ

Level 200 • First Semester • PastQuestion

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Prokaryotes has lesser functional genes, has no distinct structure
Eukaryotes has more functional genes, they are well defined (organelles), it has distinct, organized, functional structure

OSUN STATE UNIVERSITY, OSOGBO
 Department of Zoology
 FACULTY OF BASIC AND APPLIED SCIENCES,
 2018/19 HARMATTAN SEMESTER EXAMINATIONS
 COURSE CODE: BIO 203/207.
 COURSE TITLE: INTRODUCTION TO GENETICS AND CELL BIOLOGY

Instruction: Answer all questions in Section A and any other two from section B
 Time Allowed: 2 hours.

SECTION A (Answer all Questions)

- Define the following terms:
 - Species: a group of living organisms consisting of similar individuals capable of exchanging genes.
 - Alleles: it is one or two or more DNA sequences occurring at a particular gene locus.
 - Chiasmata: it is the point at which paired chromosomes remain in contact.
 - Locus: is the point/location of gene on the chromosome.
 - Synapsis: is the pairing of homologous chromosomes that occurs during meiosis.
 - Homologous Chromosomes: it is made up of chromosome pair of approximately the same length.
 - Acentric Chromosomes: it is a fragment of a chromosome that lacks centromere.
 - Haploid Chromosome Number: ...
 - Somatic Chromosome: ...
 - Centromere: it is the region that separates it into a short arm (p) and long arm (q).
- State five (5) points each on the significance of mitosis and meiosis. (5 marks)
- List six (6) differences between Mitosis and Meiosis. (6 marks)
 - Describe interphase. *replication - Duplication of DNA* (4 marks)

SECTION B (answer any two questions from this section)

- Draw diagrams to illustrate stages of Meiosis I and Meiosis II. (12 marks)
- State the result of Mitosis and Meiosis during the process of cell division. (3 marks)
- Use Punnet Square to illustrate Mendel's dihybrid experiment. (10 marks)

X (5) (a) Make labelled diagrams of a plant cell and an animal cell. (8 marks)
 (b) Distinguish between prokaryotic cell and eukaryotic cell. (6 marks)
 (c) What kind of information is contained in chromosomes? *height of human, genetic information, eye color, hair color* (1 mark)
 (d) List the four (4) basic principles of genetics that Mendel discovered in his experiments. (4 marks)
 (e) Suppose that for an organism, $2N = 24$. How many chromosomes do the characteristic such as flower colour, plant height and seed shape were controlled by pairs of heritable factors that come from diff. parents? *12*
 A factor (like dominance) could mask the presence of another.

Gene + is a unit or heredity, which performs a specific function. It is one or ~~heredity~~ more alternative form of gene.

locus - is the position/location of gene

Can you explain the difference between the two types of gametes?

Structure of DNA

(f) Distinguish among the following: Homologous chromosomes, sister chromatids, centromere, telomere, kinetochore, spindle fibers, metaphase plate, anaphase, telophase, cytokinesis, mitosis, meiosis, crossing over, independent assortment, genetic recombination, mutation, point mutation, silent mutation, missense mutation, nonsense mutation, frameshift mutation, deletion, insertion, duplication, inversion, translocation, reciprocal translocation, non-reciprocal translocation, Robertsonian translocation, Philadelphia chromosome, Down syndrome, Klinefelter syndrome, Turner syndrome, Patau syndrome, Edwards syndrome, Karyotype, pedigree chart, autosomal recessive, autosomal dominant, sex-linked recessive, sex-linked dominant, carrier, proband, consanguinity, pedigree analysis, Mendel's laws, Hardy-Weinberg equilibrium, genetic drift, natural selection, artificial selection, speciation, adaptive radiation, convergent evolution, divergent evolution, molecular clock, phylogenetic tree, cladistics, systematics, taxonomy, binomial nomenclature, genus, species, subspecies, variety, cultivar, strain, race, ethnicity, population, community, ecosystem, biome, biosphere, ecology, environmental science, conservation biology, restoration ecology, landscape ecology, urban ecology, agricultural ecology, marine ecology, freshwater ecology, terrestrial ecology, paleontology, archaeology, anthropology, linguistics, history, geography, geology, meteorology, climatology, oceanography, atmospheric science, earth science, space science, astronomy, astrophysics, cosmology, physics, chemistry, biology, medicine, dentistry, pharmacy, nursing, psychology, sociology, political science, economics, law, education, engineering, technology, computer science, information science, data science, artificial intelligence, robotics, aerospace, defense, energy, environment, agriculture, food science, nutrition, health care, sports science, recreation, tourism, culture, arts, humanities, social sciences, natural sciences, physical sciences, life sciences, earth and planetary sciences, space and atmospheric sciences, interdisciplinary sciences, transdisciplinary sciences, emerging sciences, frontier sciences, cutting-edge research, innovative research, groundbreaking research, paradigm-shifting research, transformative research, disruptive research, game-changing research, moonshot research, grand challenges research, grand challenges in science, grand challenges in technology, grand challenges in society, grand challenges in the environment, grand challenges in health, grand challenges in education, grand challenges in the economy, grand challenges in the future, grand challenges in the world, grand challenges in the universe, grand challenges in everything.

(g) How do you write an allele?

(im)

Differentiate between heritable and non-heritable variation

(b) Considering an organism with cells having diploid chromosome number of $2n$ and having two medium-sized acrocentrics and two small-sized acrocentrics.

Function of chromosomal heterocentrics. Draw side by side the following pairs of cell division stages in the cell cycle, comprising 0 two long heterocentrics, two short heterocentrics, two long heterocentrics, two short heterocentrics. Describe the major events in the individual stages and point out

• Having one representative of the organism, outside the major difference(s) between the paired stages for the transfer of information (that is, the day).

- i. Mitotic Metaphase and Metaphase I
- ii. Mitotic Anaphase and Anaphase I

- they are responsible for genetic transfer of DNA
- iv. Metaphase I and Metaphase II

in anaphase I and anaphase II

Write short notes on the following:

- Co-dominance
- Over-dominance

1. "The Mother"
2. "The Daughter"
3. "The Son"

(8) With the aid of appropriate diagram

Cells
Microtiter - is measuring
of organisms

B. Study the following sequences of the
4th and RNA sequences:

appearance - e. coli virus
 homologous - identical collection
 ATGATGTCGTTAA

identical alleles
 3-20 years - one
 century - one
 TAC TAGA GCA
 GGAAG TTTTCCCG
 CATC CAAGCAATC

c. Define transcription and translation

d. With the aid of a diagram, describe DNA template strand

transcriptum	Δ 279	258
Polonia		us conf. d. l.

Cythere, Guanine, Guanine.

Thymus.

for manuscript review. The production of mass paper

RNA polymerase: The use of mRNAs to direct protein synthesis

Part of signal processing of the protein molecule

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OSUN STATE UNIVERSITY, OSOGBO
FACULTY OF BASIC AND APPLIED SCIENCES
DEPARTMENT OF PLANT BIOLOGY
2019/2020 HANAIITARN SEMESTER EXAMINATIONS
COURSE CODE: BIO 203
COURSE TITLE: INTRODUCTORY GENETICS AND CELL BIOLOGY
NUMBER OF UNITS: 2
TIME ALLOWED: 2 HOURS

SECTION A
INSTRUCTION: Answer All Questions in this section

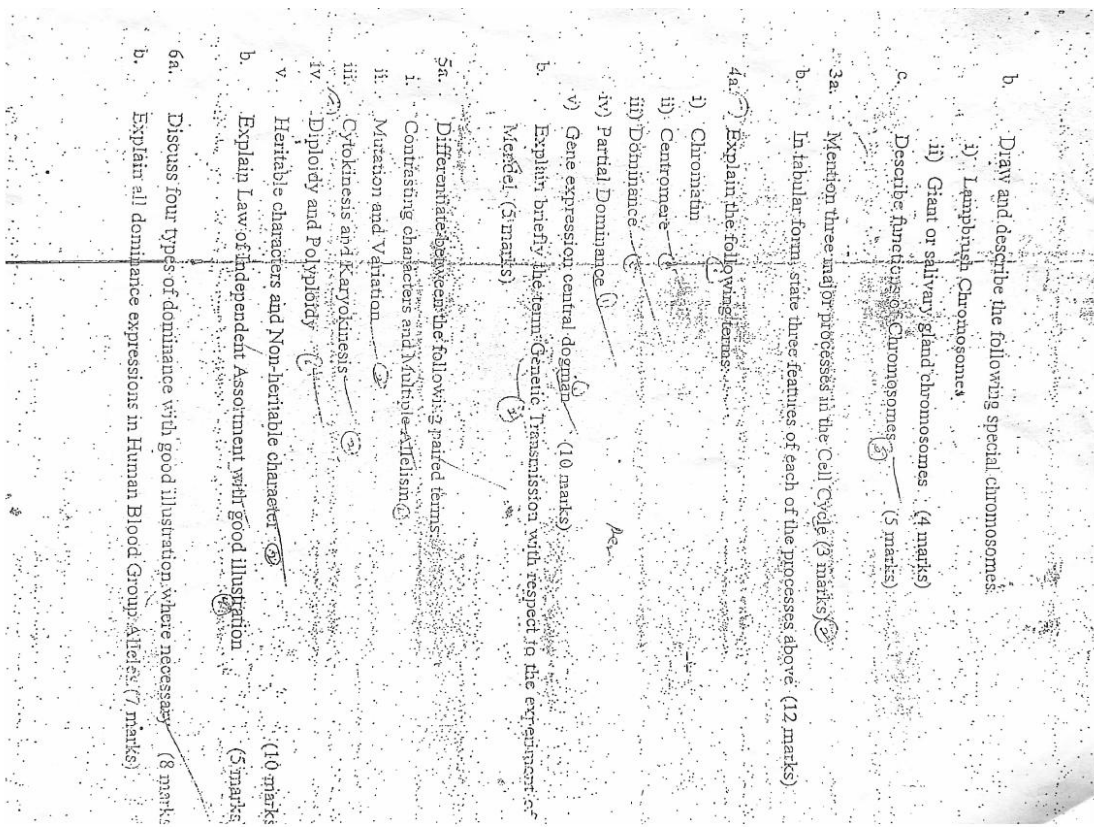
1a. Draw, label and describe a typical chromosome (5 marks)
b. Define the following genetic terms:
i. Genotype
ii. Phenotype
iii. Karyotype
iv. Locus
v. Homologous Chromosomes
vi. Saturated Chromosomes
vii. Acentric Chromosomes
viii. Dicentric Chromosomes
ix. Acrocentric Chromosomes
x. Alleles

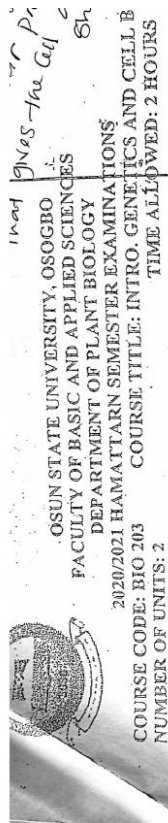
c. State the characteristics of heritable variation (5 marks)
d. State the characteristics of sexual reproduction (5 marks)

SECTION B
INSTRUCTION: Answer any THREE questions from section B

2a. Describe the variations in chromosome using the following criteria:
i. Size
ii. Number of centromere
iii. Position of centromere (6 marks)

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SECTION A: Answer all questions in this section

1/A. Define the following terms:

- a) Cytokinesis ✓
- b) Homologous chromosomes ✓
- c) Genotype ✓
- d) Haploid chromosome number ✓
- e) Locus ✓
- f) Karyokinesis ✓
- g) Kinetochore
- h) Asynapsis
- i) Chiasmata (9 marks)
- (3 marks)
- (3 marks)

B. State three (3) characteristics of Heritable variations

C. State three characteristics of sexual reproduction

2a. Draw a typical chromosome and state the function of any three (3) parts of it

(5 marks)

b. State two different types of plasmids and their functions

c. Differentiate between the following pair of terms:

i) Gene and locus

ii) RNA and DNA

iii) Cell elongation and cell differentiation

(6 marks)

SECTION B: Answer any two questions in this section

3a. Define Meiosis

b. What is a meiocyte?

c. What are the end products of meiosis?

d. With the aid of good diagrams, describe the chromosomal activities and events that take place during the second, third and fifth sub-stages of Prophase I in an organism with somatic chromosome number of $2n = 6$.

(10 marks)

- 4a. What is Centromere?
 - b. What is the name given to chromosomes without centromere?
 - c. Describe the various types of chromosomes based on the number of centromere(s) on them.
 - d. Draw and describe the various types of monocentric chromosomes (10 marks)
- 5a. In which part of the cell does the genetic materials reside?
- 4b. Describe the experiment to support your answer in "a" above, using *Acetabularia* as proven by Hammerling (1892). (10 marks)

SECTION C: Answer any two questions in this section

- 6a. Explain briefly the hierarchies of chromosomal organization
 - b. State five functions of chromosome
 - c. Describe histones and their roles in chromosomes (10 marks)
- 7a. Explain briefly organization of chromosome in Bacteria
- b. Differentiate between heterochromatin and euchromatin
- c. Write the genetic codes and their respective pairing patterns (10 marks)
- 8a. Describe gene expression with an illustrative equation
- b. List the four steps involved in the process of Transcription
- c. Explain briefly and with appropriate symbols the Mendelian Law of Dominance (10 marks)
- 9a. Name the scientists that proposed the cell theory. *Handwritten: Schwann, Schleiden, Virchow*
- b. Write the major ideas explained in the cell theory
- d. State four major differences between prokaryotes and Eukaryotes
- e. Describe the shape of the following cells:
(i) Coccus (ii) Amoeba (iii) Paramecium (iv) Bacillus (10 marks)

OSUN STATE UNIVERSITY, OSOGBO
FACULTY OF BASIC AND APPLIED SCIENCES
DEPARTMENT OF PLANT BIOLOGY
2019/2020 FARMATTAIN SEMESTER EXAMINATIONS
COURSE CODE: BIO 203
COURSE TITLE: INTRODUCTORY GENETICS AND CELL BIOLOGY
NUMBER OF UNITS: 2
TIME ALLOWED: 2 HOURS

SECTION A
INSTRUCTION: Answer All Questions in this section

1a. Draw, label and describe a typical chromosome (5 marks)

b. Define the following genetic terms:

- Genotype
- Phenotype
- Karyotype
- Locus
- Homologous Chromosomes
- Satellited Chromosomes
- Acentric Chromosomes
- Dicentric Chromosomes
- Acrocentric Chromosomes
- Alleles

(10 marks)

c. State the characteristics of heritable variation (5 marks)

d. State the characteristics of sexual reproduction (5 marks)

SECTION B
INSTRUCTION: Answer any THREE questions from section B

2a. Describe the variations in chromosome using the following criteria:

- Size
- Number of centromere
- Position of centromere

(6 marks)





OSUN STATE UNIVERSITY, OSOGBO
College of Science, Engineering and Technology
Faculty of Basic and Applied Sciences
Department of Zoology
2021/2022 HARMATTAN SEMESTER EXAMINATION

PROGRAMME: B. Sc (Zoology) COURSE TITLE: Cell Biology and Genetics I
COURSE CODE: BIO 203/207 NO OF UNITS: 2
TIME ALLOWED: 2 hours

INSTRUCTION: Answer all questions in Section A and four questions from section B

SECTION A

1. (a) Explain the following terms:

- (i) Chiasmata
- (ii) Locus
- (iii) Lysosome
- (iv) Mitochondrion
- (v) Cytoskeleton
- (vi) Test Cross

b) Name each stage of mitosis described

- (i) Chromosomes line up in the equatorial plane
- (ii) Nuclear membrane reforms and cytokinesis occurs
- (iii) Chromosomes become visible, spindle apparatus forms
- (iv) Sister chromatids move to opposite poles of cell

(4 x 1/2 = 2 marks)

c) Enumerate the major ideas of the cell theory (2 marks)

d) Distinguish between the following pair of alleles:

- (i) Dominant allele and recessive allele
- (ii) Genotypes and phenotype
- (iii) Cytokinesis and karyokinesis

(3 marks)

e) State and explain any three types of aneuploidy of chromosome variation. (2 marks)

2. (a) Draw and label appropriately a typical chromosome (5 marks)

b) Describe organization of chromosome in a named prokaryote and eukaryote (6 marks)

c) State the functions of chromosomes in living cells (4 marks)

metaphase
anaphase
telophase
cytokinesis

SECTION B

3 (a) State the second law of inheritance

(b) There are two alleles for hair colour, black and white and two alleles for hair length, short and long. In a breeding experiment all the F_1 phenotype produced from a cross between pure breeding short black hair and a pure breeding long white hair parents had short black hair.

- (i) Which alleles are dominant?
- (ii) State the expected proportion on F_2 phenotype and genotype
- (iii) What proportion of F_2 will be short and black?
- (iv) What proportion of F_2 will be homozygous recessive for both loci
- (v) What proportion of F_2 will be heterozygous for both loci. (10marks)

4 (a) State the first law of inheritance

(b) In a variegated plant, the shape of the leaf margin is either crenate margin or deep lobed margin. The less common deep lobed margin leaf has dominance gene.

- (i) What are the phenotypic and genotypic results of the cross?
- (ii) What is the probability that F_2 will produce a dominant phenotype among the progeny?
- (iii) What is the probability that of picking a deep lobed margin plant from the progeny of F_2
- (iv) What is the probability that the deep lobed will be heterozygote (10marks)

5 (a) Draw illustrative diagram of cell cycle and explain its phases (5 marks)

b) Explain the following processes in living cell: (i) Cell division (ii) Cell enlargement (iii) Cell differentiation (iv) Nuclear inheritance (v) Allelic dominance (5 marks)

6 (a) What are plasmids? (2 marks)

b) Mention types of plasmid and their functions (5 marks)

c) Explain the following terms: (i) Nucleoids (ii) Nucleosome (iii) Chromatin fiber (iv) Heterochromatin (v) Scaffolds (vi) Condensation (3 marks)

SECTION B continues

7. (a) Write two differences between prokaryotic and eukaryotic cells (2marks).
(b) Make a large labelled diagram of a typical animal cell (5marks).
(c) Describe the following terms associated with plant cells:
Cell wall (ii) Plastids (iii) Vacuole (3marks).
8. (a) What is gametogenesis? (1mark).
(b) Discuss the process of egg formation (oogenesis) in higher animals (5marks).
(c) Briefly describe Weismann's experiment that disproved the popular belief of inheritance of acquired character and his eventual theory of germplasm (4marks)

FACULTY OF BASIC AND APPLIED SCIENCES
DEPARTMENT OF BIOLOGICAL SCIENCES

HAROLDIAN SEMESTER EXAMINATIONS
COURSE CODE: BIO 203
SESSION: 2016/2017
COURSE TITLE: MOLOGY
TIME: 2 HOURS

LATE REGISTRATION
INSTRUCTIONS: Answer two questions from each section.

SECTION A

1. (a) What are structural genes? Describe concisely the different components of structural genes. (10 marks)

(b) Define gene regulation and highlight mechanisms by which it occurred. (10 marks)

(c) Distinguish between Mendelian and Non-Mendelian traits. (10 marks)

(d) Describe briefly the stages involved in the processes of gene expression. (20 marks)

2. (a) Use Punnett square to predict the genotypes and phenotypes of the first filial (F₁) and second filial (F₂) generations of offspring in a monohybrid test cross involving homozygous pea plants with grey seed coat colour (GG) and white seed coat colour (gg). (10 marks)

(b) Give the interpretation of the Punnett square results for the F₁ and F₂ crossing. (10 marks)

(c) Describe the phenomenon that happened to white seed coat colour and explain it. (10 marks)

(d) Define the following: (i) polycyclic traits and (ii) polycyclic traits. (10 marks)

(e) Define the following: (i) polycyclic traits and (ii) polycyclic traits. (10 marks)

(f) Highlight the processes involved in the synthesis of protein from deoxyribonucleic acid (DNA). (20 marks)

SECTION B

1. (a) State the contributions of the following scientists in the history of the discovery of cell. (10 marks)

(b) (i) Robert Hooke and (ii) Robert Brown. (10 marks)

2. (a) With the aid of a schematic diagram only, describe the process of mitosis. (9 marks)

(b) Using a well-labelled diagram, describe the morphology of a chromosome. (12 marks)

3. (a) State the cell theory. (3 marks)

(b) Explain the structure of cell membrane using the fluid mosaic model. (10 marks)

(c) With the aid of a diagram, differentiate between types of human chromosome. (8 marks)

(d) State the differences and functions of smooth and rough endoplasmic reticulum. (3 marks)


OSUN STATE UNIVERSITY, OSOGBO
FACULTY OF BASIC AND APPLIED SCIENCES
DEPARTMENT OF PLANT BIOLOGY
2020/2021 HAMATTARN SEMESTER EXAMINATIONS
COURSE CODE: BIO 203 **COURSE TITLE: INTRO. GENETICS AND CELL BIO.**
NUMBER OF UNITS: 2 **TIME ALLOWED: 2 HOURS**

SECTION A: Answer all questions in this section

1 A. Define the following terms:

a) Cytokinesis d) Haploid chromosome number g) Kinetochore	b) Homologous chromosomes e) Locus h) Anaphase
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c) Genotype
f) Karyokinesis
i) Chiasmata

(9 marks)
(3 marks)
(3 marks)

B. State three (3) characteristics of Heritable variations

C. State three characteristics of sexual reproduction

2a. Draw a typical chromosome and state the function of any three (3) parts of it

(5 marks)
(4 marks)

b. State two different types of plasmids and their functions

c. Differentiate between the following pair of terms:

i) Gene and locus ii) RNA and DNA iii) Cell elongation and cell differentiation	(6 marks)
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SECTION B: Answer any two questions in this section

3a. Define Meiosis

b. What is a meiocyte?

c. What are the end products of meiosis?

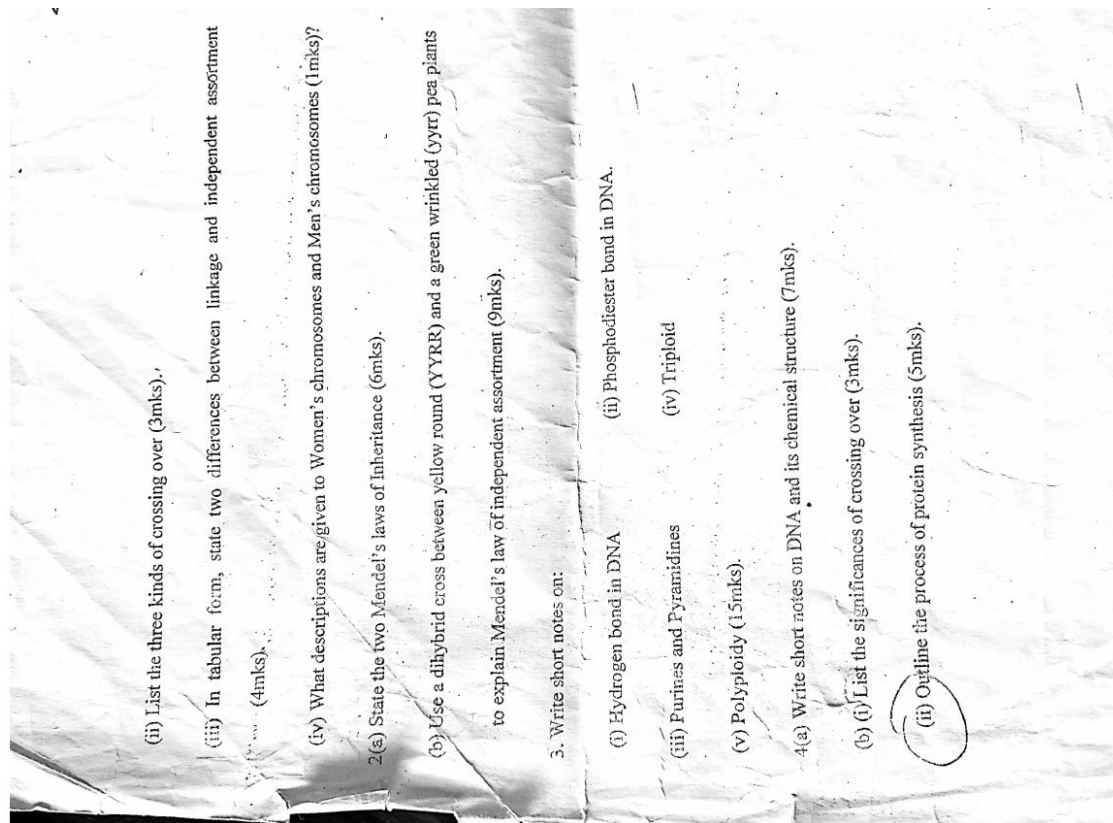
d. With the aid of good diagrams, describe the chromosomal activities and events that take place during the second, third and fifth sub-stages of Prophase I in an organism with a somatic chromosome number of $2n = 6$.

(10 marks)

- 4a. What is Centromere?
- b. What is the name given to chromosomes without centromere? Acentric chromosome
- c. Describe the various types of chromosomes based on the number of centromere(s) present on them.
Monocentric
Policentric
Acentric
- d. Draw and describe the various types of monocentric chromosomes (10 marks)
- 5a. In which part of the cell does the genetic materials reside? Nucleus in the nucleus
- b. Describe the experiment to support your answer in "a" above, using *Acerabularia* species as proven by Hammerling (1892) (10 marks)

SECTION C: Answer any two questions in this section

- 6a. Explain briefly the hierarchies of chromosomal organization
- b. State five functions of chromosome
- c. Describe histones and their roles in chromosomes (10 marks)
- 7a. Explain briefly organization of chromosome in Bacteria
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- 8a. Describe gene expression with an illustrative equation
- b. List the four steps involved in the process of Transcription
- c. Explain briefly and with appropriate symbols the Mendelian Law of Dominance
- 9a. Name the scientists that proposed the cell theory. Mary Schlegel, Robert Brown, Schleiden, Schwann (10 marks)
- b. Write the major ideas explained in the cell theory
- d. State four major differences between prokaryotes and Eukaryotes
- e. Describe the shape of the following cells:
 (i) Coccus (ii) Amoeba (iii) Paramecium (iv) Bacillus (10 marks)



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